

A Short Exploratory Note on a Convexity Step in Theorem 11

Context and purpose. While reading Theorem 11 of *Characterizing Online and Private Learnability under Distributional Constraints via Generalized Smoothness*, I tried to trace the necessity direction for private learnability under a general distribution class $\mathcal{U}$. The statement of Theorem 11 appears to be formulated for a general distribution class $\mathcal{U}$, while my reading of the lower-bound route in Appendix F.1 led me to Lemma 18 and Theorem 19, where convexity of $\mathcal{U}$ is assumed.

The purpose of this note is not to claim that the theorem is incorrect. Rather, I describe the proof path as I currently understand it, record one convexification check, and isolate the step where I do not yet see the bridge from the general statement to the convex reduction.

1 The proof route and the convexity point

The proof in Appendix F.1 seems to argue by contrapositive. If $\mathcal{U}$ is not generalized smooth, then the envelope submeasure

$$\mu_{\mathcal{U}}(A) := \sup_{\mu \in \mathcal{U}} \mu(A)$$

is not continuous. In this framework, this implies that there exists a scale $\eta > 0$ such that the fragmentation number

$$N_{\mathcal{U}}(\eta)$$

is infinite. The proof then invokes Theorem 19 to obtain a private-learning impossibility result for generalized threshold classes.

The step I am trying to understand is therefore

$$N_{\mathcal{U}}(\eta) = \infty \implies \text{private-learning impossibility.}$$

My uncertainty is that Theorem 19 is stated for a convex distribution class, whereas Theorem 11 itself does not appear to include convexity as an assumption.

This is also the point where I see convexity entering Lemma 18. In that proof, one chooses distributions

$$\mu_1, \dots, \mu_K \in \mathcal{U}$$

and forms a mixture

$$\mu = \sum_{i=1}^K \nu_i \mu_i.$$

The argument then uses that $\mu \in \mathcal{U}$, which is immediate when $\mathcal{U}$ is convex. This matters because the private learner is only assumed to work for distributions whose marginal lies in $\mathcal{U}$. If the constructed mixture is outside $\mathcal{U}$, it is not clear that the learner promised by the theorem can be applied to it.

So my current reading is that Lemma 18 and Theorem 19 establish the lower-bound route for convex $\mathcal{U}$, and I am trying to understand how this route transfers to the general $\mathcal{U}$ appearing in Theorem 11.

2 Convexification preserves the structural obstruction

A natural attempt is to replace $\mathcal{U}$ by its convex hull $\text{conv}(\mathcal{U})$, apply Theorem 19 there, and then transfer the conclusion back to $\mathcal{U}$.

At the level of the envelope, this works cleanly for the finite convex hull. For every measurable set A ,

$$\mu_{\text{conv}(\mathcal{U})}(A) = \sup_{\bar{\mu} \in \text{conv}(\mathcal{U})} \bar{\mu}(A) = \sup_{\mu \in \mathcal{U}} \mu(A) = \mu_{\mathcal{U}}(A).$$

Indeed, $\bar{\mu}(A)$ is only a convex combination of values $\mu(A)$, so convexification cannot increase the supremum. Hence convexification preserves the envelope, and therefore also preserves the fragmentation numbers:

$$N_{\text{conv}(\mathcal{U})}(\eta) = N_{\mathcal{U}}(\eta).$$

Thus, from the structural side, convexification does not change the obstruction. The remaining question is whether the private-learning assumption in Theorem 11 also passes from $\mathcal{U}$ to $\text{conv}(\mathcal{U})$. I do not currently see why this is automatic.

3 Why the learning guarantee does not seem to lift immediately

As a small sanity check, I tested the proposed lifting on the simplest two-component case

$$\mathcal{U} = \{\mu_1, \mu_2\}.$$

Then $\text{conv}(\mathcal{U})$ contains

$$\bar{\mu} = \lambda\mu_1 + (1 - \lambda)\mu_2.$$

Suppose similarly that

$$\bar{D} = \lambda D_1 + (1 - \lambda)D_2,$$

where the X -marginals of D_1, D_2 are μ_1, μ_2 . An m -sample from $\bar{D}$ has law

$$\bar{D}^m = (\lambda D_1 + (1 - \lambda)D_2)^m.$$

Expanding the product gives

$$\bar{D}^m = \sum_{j_1, \dots, j_m \in \{1, 2\}} \lambda^{|\{t: j_t=1\}|} (1 - \lambda)^{|\{t: j_t=2\}|} D_{j_1} \otimes \dots \otimes D_{j_m}.$$

This is not the same as drawing one latent component for the whole dataset, namely

$$\lambda D_1^m + (1 - \lambda)D_2^m.$$

This distinction seems relevant because the assumed guarantee under $\mathcal{U}$ controls the learner on samples from D_1^m and D_2^m separately. It does not directly control the heterogeneous product terms

$$D_{j_1} \otimes \dots \otimes D_{j_m},$$

where different coordinates of the same sample may come from different mixture components.

Equivalently, the desired guarantee under $\bar{D}$ involves

$$\mathbb{E}_{S \sim \bar{D}^m} [\text{err}_{\bar{D}}(A(S))].$$

Although the error of a fixed hypothesis is linear in the data distribution, the output $A(S)$ is data-dependent. Therefore, I do not see an immediate argument showing that guarantees under D_1^m and D_2^m imply the corresponding guarantee under $\bar{D}^m$.

Of course, this does not rule out such a lifting argument; it only explains why I do not see it as an immediate consequence of linearity.

4 Findings and remaining question

The point I am trying to isolate occurs before the known private threshold lower bound is applied: it is the reduction step that turns a learner under $\mathcal{U}$ into a learner for thresholds on the constructed finite domain.

Based on the checks above, my current understanding is:

- Convexification preserves the envelope submeasure and the fragmentation obstruction.
- The lower-bound reduction in Lemma 18 and Theorem 19 appears to use convexity to keep constructed mixtures inside $\mathcal{U}$.
- It is not clear to me that private learnability over $\mathcal{U}$ automatically extends to private learnability over $\text{conv}(\mathcal{U})$, because product samples from mixture distributions contain heterogeneous component assignments.
- The issue seems to occur before the known private threshold lower bound is applied, namely at the step where one passes from a learner under $\mathcal{U}$ to a learner for thresholds on the constructed finite domain.

This leaves me with the following question.

I am trying to understand whether the proof of Theorem 11 uses an implicit convexification or lifting step from $\mathcal{U}$ to $\text{conv}(\mathcal{U})$, or whether the private necessity direction should instead be interpreted directly at the level of the envelope submeasure.

More concretely, is there a standard argument showing that the lower-bound route through Lemma 18 and Theorem 19 transfers from convex distribution classes to the general class appearing in Theorem 11?

Based on these checks, my current understanding is that the structural obstruction survives convexification, while the private-learning guarantee seems to require an additional lifting argument. I may simply be missing a standard argument, and I would be very grateful for any correction or guidance on how this step should be understood.

References

- [1] M. Blanchard, A. Shetty, and A. Rakhlin. *Characterizing Online and Private Learnability under Distributional Constraints via Generalized Smoothness*. arXiv:2602.20585, 2026.
- [2] N. Alon, R. Livni, M. Malliaris, and S. Moran. Private PAC learning implies finite Littlestone dimension. *Proceedings of the 51st Annual ACM Symposium on Theory of Computing*, 2019.